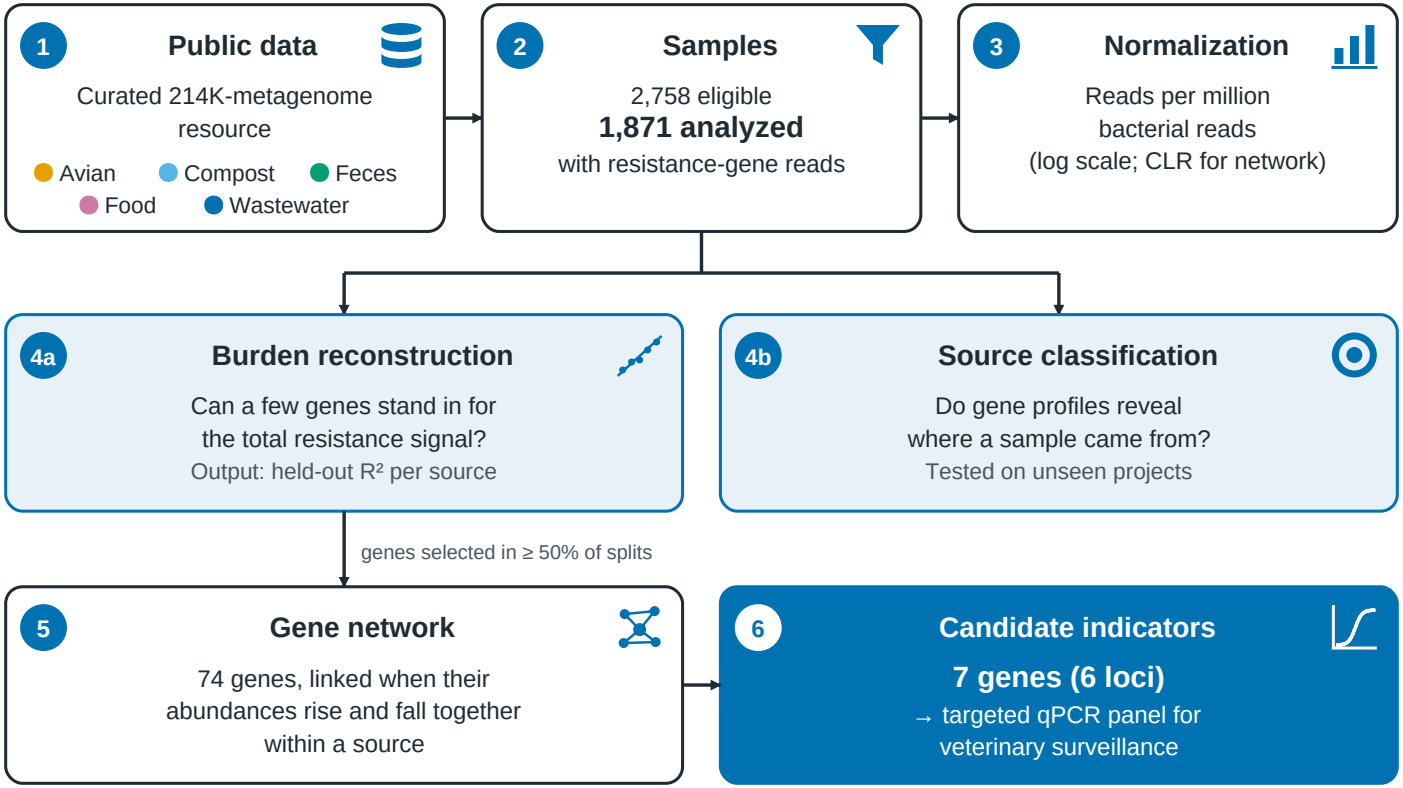
1
Public data
Curated 214K-metagenome resource
Avian
Compost
Feces
Food
Wastewater
2
Samples
2,758 eligible
1,871 analyzed
with resistance-gene reads
3
Normalization
Reads per million bacterial reads
(log scale; CLR for network)
4a
Burden reconstruction
Can a few genes stand in for the total resistance signal?
Output: held-out R² per source
4b
Source classification
Do gene profiles reveal where a sample came from?
Tested on unseen projects
genes selected in ≥ 50% of splits
5
Gene network
74 genes, linked when their abundances rise and fall together within a source
6
Candidate indicators
7 genes (6 loci)
→ targeted qPCR panel for veterinary surveillance